

Generation Interconnection System Impact Study Report

For Queue Project Q22-2061

Asarco 138 kV
December 31, 2023

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2061. The Interconnection Customer proposes a 132.9 MW solar photovoltaic facility interconnecting at the Asarco 138 kV substation in the Kentucky area. The request was received on April 10, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 551 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 21 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$29,022,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Sample Rd / Tomahawk-Atc Substation 138.0 kV	Latah Junction / Spatsum Lornex 138.0 kV L	155.8	225	23.2
Welfare / Kettle 138.0 kV Line	Spatsum Lornex / Latah Jct 138.0 kV line	153.2	256	23.2
Alibates Substation / Latah Jct 138.0 kV	Harvest Power Orlando / Frontier 138.0 kV line	148.9	495	45.0
Latah Junction / Spatsum Lornex 138.0 kV line	Frontier / Harvest Power Orlando 138.0 kV	147.4	508	45.0
Monrovia / Sample Road 138.0 kV	Latah Junction Substation / Spatsum Lornex 138.0 kV	146.3	230	25.5
Kettle / Country Club 2 138.0 kV	Spatsum Lornex / Latah Jct 138.0 kV	142.3	347	25.1
Welfae / Tomahawk-Atc 138.0 kV Line	Latah Jct / Spatsum Lornex 138.0 kV	141.3	265	23.2
Country Club 2 / Spatsum Lornex 138.0 kV	Latah Junction / Sptsum Lornex 138.0 kV	141.0	355	25.1
Harvest Power Orlando / E Broad Substation 138.0 kV	Spatsum Lornex / Latah Junction 138.0 kV	133.8	278	27.0
Harvest Power Orlando / Frontier 138.0 kV	Latah Junction / Spatsum Lornex Substation 138.0 kV	128.5	488	41.9
Alibates / Juliette 138.0 kV Line	Alibates / Action 138.0 kV line	121.9	368	26.0
N City / Boonville Substation 138.0 kV	Spatsum Lornex Substation / Latah Jct 138.0 kV	120.3	195	14.0
Boonville / Sugar Crk Power 138.0 kV	Spatsum Lornex / Latah Junction 138.0 kV	110.1	221	14.0
Yuba City Sta A / Frontier 138.0 kV	Spatsum Lornex / McClellan Ar 138.0 kV L	109.7	586	61.0

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Yuba City Sta A / Westin-Elgin 138.0 kV	Spatsum Lornex / McClellan Ar 138.0 kV	109.3	600	61.0
Eldorado Substation / Action 138.0 kV	Pine Bluff Watson Chapel / Carrazo 1201 345.0 kV	107.4	751	37.5
Alibates / Action 138.0 kV	Carraizo 1201 Substation / Pine Bluff Waton Chapel 345.0 kV	106.3	798	37.5
McClellan Ar / Spatsum Lornex 138.0 kV	Westin-Elgin / Yuba City Sta A 138.0 kV	105.1	632	64.5
McClellan Ar / Westin-Elgin 138.0 kV	Yuba City Sta A / Westin-Elin 138.0 kV	104.0	502	47.4
Delano / Kettle 138.0 kV	Yuba City Sta A / Westin-Elin 138.0 kV line	102.0	194	12.1
S / Cortez 138.0 kV	Summit Lake Wps / Edna 138.0 kV	101.5	226	11.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sample Rd / Tomahawk-Atc Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$10,330,000	16% (\$1,597,000)
Welfare / Kettle 138.0 kV Line	Rebuild line with higher-capacity conductor	\$12,161,000	16% (\$1,880,000)
Alibates Substation / Latah Jct 138.0 kV	Rebuild line with higher-capacity conductor	\$13,829,000	15% (\$2,069,000)
Spatsum Lornex / Latah Junction 138.0 kV line	Rebuild line with higher-capacity conductor	\$6,918,000	15% (\$1,035,000)
Monrovia / Sample Road 138.0 kV	Rebuild line with higher-capacity conductor	\$17,744,000	15% (\$2,698,000)
Kettle / Country Club 2 138.0 kV	Rebuild line with higher-capacity conductor	\$6,286,000	14% (\$893,000)
Welfae / Tomahawk-Atc 138.0 kV Line	Rebuild line with higher-capacity conductor	\$6,246,000	16% (\$966,000)
Spatsum Lornex / Country Club 2 138.0 kV	Rebuild line with higher-capacity conductor	\$12,965,000	14% (\$1,841,000)
Harvest Power Orlando / E Broad Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$11,686,000	16% (\$1,848,000)
Harvest Power Orlando / Frontier 138.0 kV	Rebuild line with higher-capacity conductor	\$28,610,000	16% (\$4,538,000)
Alibates / Juliette 138.0 kV	Rebuild line with higher-capacity conductor	\$12,831,000	15% (\$1,880,000)
N City / Boonville 138.0 kV	Rebuild line with higher-capacity conductor	\$15,460,000	13% (\$1,985,000)
Boonville / Sugar Crk Power 138.0 kV	Rebuild line with higher-capacity conductor	\$4,912,000	13% (\$631,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Yuba City Sta A / Frontier 138.0 kV	Reconductor line	\$3,761,000	20% (\$767,000)
Yuba City Sta A / Westin-Elgin 138.0 kV	Reconductor line	\$8,275,000	20% (\$1,686,000)
Eldorado / Action 138.0 kV	Reconductor line	\$629,000	7% (\$44,000)
Alibates / Action 138.0 kV	Reconductor line	\$6,127,000	7% (\$424,000)
Spatsum Lornex / Mcclellan Ar 138.0 kV Line	Reconductor line	\$4,687,000	20% (\$923,000)
Mcclellan Ar / Westin-Elgin 138.0 kV	Reconductor line	\$3,484,000	20% (\$686,000)
Delano / Kettle 138.0 kV	Reconductor line	\$3,670,000	11% (\$393,000)
S / Cortez 138.0 kV	Reconductor line	\$2,121,000	11% (\$238,000)
	Total allocated to project		\$29,022,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.